

Zoe Hackshaw

Curriculum Vitae

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EDUCATION	Ph.D. in Astronomy , The University of Texas at Austin Advisor: Dr. Keith Hawkins Expected June 2027	
	M.A. in Astronomy , The University of Texas at Austin Thesis: <i>Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE</i> Advisor: Dr. Keith Hawkins	Dec. 2024
	B.S. in Astrophysics , The University of Florida <i>summa cum laude</i> Thesis: <i>J0010-1735—A Stellar Journey from an R-process Enhanced Ultra Faint Dwarf Galaxy to the Milky Way Halo</i> Advisor: Dr. Rana Ezzeddine	May 2021
APPOINTMENTS	Graduate Student Researcher Department of Astronomy, The University of Texas at Austin Advisor: Dr. Keith Hawkins	2022 - Present
	Pre-Doctoral Fellow Center for Computational Astrophysics, Flatiron Institute Advisors: Dr. Adrian Price-Whelan, Dr. Carrie Filion, Dr. Aida Behmard	Jan. 2026 - May 2026
	Undergraduate Student Researcher Department of Astronomy, The University of Florida Dr. Rana Ezzeddine	2019 - 2021
SELECT HONORS AND AWARDS	Center for Computational Astrophysics Pre-Doctoral Fellowship Flatiron Institute, Simons Foundation	2026
	David L. Lambert Graduate Fellowship Department of Astronomy, The University of Texas at Austin	2024
	Senior Thesis Highest Honors Department of Astronomy, The University of Florida	2021
PUBLICATIONS	First Author	
	🖱️ Clickable link 2. <i>A Detailed Chemical Analysis of the Red Giant Orbiting Gaia BH3: From Lithium to Thorium</i> Hackshaw Z. , Hawkins K., and Manea C. 2025 ApJ , DOI 10.3847/1538-4357/ae1617	
	1. <i>[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE</i> Hackshaw Z. , Hawkins K., Filion C., Horta D., Laporte C. F. P., Carr C., Price-Whelan A. M et al 2024 ApJ , DOI 10.3847/1538-4357/ad900e	

Contributing Author

6. *An Analytic Model for Stellar Metallicity Gradient Residuals in Cold, Phase-Mixed Galactic Disks*
Modak S., **Hackshaw Z.**, and Hawkins K. 2026, **ApJL**, DOI 10.48550/arXiv.2606.14850

5. *Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from Kinematic Ridges with Gaia*
Jurado C., Hawkins K., Hunt J. A. s., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/ arXiv.2601.02494

4. *Exploring the History of Stellar Mergers with Chemistry: Examining the Origins of Massive α -enriched Stars Using Carbon Isotope Ratios*
Maas Z. G., Hawkins K., Gerber J. M., **Hackshaw Z.**, Manea C., et al. 2025, **ApJ**, DOI 10.48550/arXiv.2509.08900

3. *Optical Spectroscopy Reveals Hidden Neutron-capture Elemental Abundance Differences among APOGEE-identified Chemical Doppelgängers*
Manea C., Ness M., Hawkins K., Zeimann G., Hogg D. W., Filion C., Griffith E., Johnston K., Casey A., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/arXiv.2508.16717

2. *Discovery of a Metal-poor Red Giant Star with the Highest Ultralithium Enhancement*
Kowkabany J., Ezzeddine R., Charbonnel C., Roederer I. U., Wang E. X., Li Y., **Hackshaw Z.**, et al 2024, **ApJ**, DOI 10.3847/1538-4357/ad6004/ arXiv.2209.02184

1. *Uranium Abundances and Ages of r-process Enhanced Stars with Novel U II Lines*
Shah S., Ezzeddine R., Ji A., Hansen T., Catelan M., **Hackshaw Z.**, et al 2023, **ApJ**, DOI 10.3847/1538-4357/acb8af

**SELECT
SCIENTIFIC
PRESENTATIONS**
👉 Clickable link

Cool Stars, Stellar Systems and the Sun 23, Poster University of Tokyo, ABC, ISAS-JAXA	June 2026
Pre-Doctoral Symposium, Talk Center for Computational Astrophysics; Flatiron Institute Columbia Univeristy	May 2026
s,i,r- Element Nucleosynthesis (sirEN) Conference, Poster Italian National Institute for Astrophysics	June 2025
Milky Way Assembly Tale Conference, Poster Istituto Nazionale Di Astrofisica	June 2024
SDSS-V Collaboration Meeting, Contributing Talk Sloan Digital Sky Survey-V	Aug. 2023
Stellar Properties from NLTE Radiative Transfer, Contributing Talk IReNA, Kavli Institute for Cosmological Physics, and the University of Chicago	May 2023
AAS Meeting, Poster American Astronomical Society	Jan. 2021

**SELECT
OBSERVING
EXPERIENCE**

PI, <i>Pollution of FGK White Dwarf Companions</i> , 7 nights Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory	Oct. 2024
PI, <i>Chemical Analysis of Gaia BH3</i> , 7 nights	June 2024

	<i>Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory</i> PI, Cosmochronometry with Asteroseismic Ages, 7 nights Feb. 2024 <i>Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory</i>
COMPUTER SKILLS	Languages: Python, Julia, C++ Software: Turbospectrum, BACCHUS, Spectroscopy Made Hard, Git, GitHub, LaTeX
TEACHING & MENTORSHIP	Graduate-Undergraduate Mentorship in astronoMY Mentor 2024 - Present <i>The University of Texas at Austin</i> Research Mentor, Students: Abigail Skelton and Nachiket Yadav 2024 <i>The University of Texas at Austin</i> Teaching Assistant, AST 325K: Stellar Astronomy 2023 <i>The University of Texas at Austin</i>
OUTREACH & SCIENCE COMM.	AAS Journal Author Series, Invited Talk June 2026 <i>Davidson Academy Online</i> Astronomy Club Guest Lecture, Invited Talk March 2026 <i>Davidson Academy Online</i> Fourth Grade Astronomy Lesson, Invited Talk Feb. 2026 <i>Eagle Ridge Elementary</i> Live Deep Sky Tour Stream, Invited Talk Dec. 2025 <i>McDonald Observatory</i> Future in STEM Career Seminar, Invited Talk Oct. 2025 <i>Lynn University</i> Astronomy Day Distance Learning Presentation, Invited Talk Sept. 2025 <i>McDonald Observatory</i> Explore our Solar System Teacher Workshop, Invited Co-Lead July 2025 <i>McDonald Observatory</i> Astronomy on Tap ATX Presentation, Contributing Talk Feb. 2025 <i>Astronomy on Tap Austin, TX</i> UT STEM Girl Day, Contributing Activity Lead Feb. 2023 - 2025 <i>The University of Texas at Austin</i> Astronomy at the Library Presentation, Contributing Talk Nov. 2024 <i>Williamson County Astronomy Club</i> Board of Visitors Summer Meeting, Invited Talk July 2024 <i>McDonald Observatory and the University of Texas at Austin</i> Girl Scout Space Badge Presentation, Invited Talk April 2022 <i>Daisy Troop 41119, Austin, TX</i>
LEADERSHIP & EXTERNAL ACTIVITIES	Social Media and Marketing Manager 2023 - Present <i>Astronomy on Tap, Austin, TX Chapter</i> Prospective Graduate Student Visit Lead 2023-2025 <i>Department of Astronomy, The University of Texas</i> Assistant Director of Mini Marathons 2020 - 2021 <i>Morale Team for Dance Marathon at the University of Florida</i> Vice President of Chapter Relations and Standards 2018 - 2019 <i>Alpha Chi Omega, Gamma Iota Chapter</i>

**PROFESSIONAL
EXPERIENCE**

Engineering Scientist Associate

Applied Research Laboratories, University of Texas at Austin

2021 - 2022